

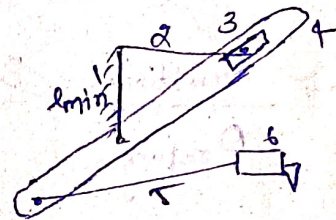
$$QRR = \frac{Q_{cutting}}{Q_{return}} > 1$$

$$\frac{Q_{cutting}}{Q_{return}} = 2 \quad \frac{Q_{return}}{Q_{cutting}} = \frac{1}{2}$$

$$QRR = \frac{360 - \beta}{\beta} = 2 \Rightarrow 360 - \beta = 2\beta \Rightarrow \boxed{\beta = 120}$$

V. Imp
 * Whitworth Quick return mechanism

$$\cos \frac{\beta}{2} = \frac{\text{length of fixed link } (l_{min})}{\text{length of cranks}}$$

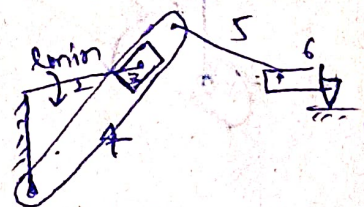


$$\cos \left(\frac{120}{2} \right) = \frac{1}{2} = \frac{\text{length of fixed link } (l_{min})}{\text{length of cranks}} = \frac{1}{QRR}$$

$$QRR = 2$$

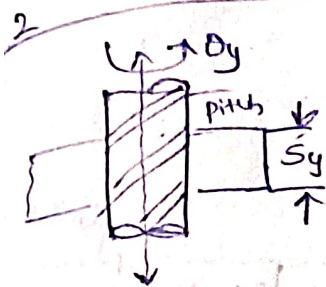
V. Imp
 * Crank slotted mechanism

$$\cos \frac{\beta}{2} = \frac{\text{length of cranks } (l_{min})}{\text{length of fixed link}}$$



$$\cos \frac{120}{2} = \frac{1}{2} = \frac{\text{length of cranks } (l_{min})}{\text{length of fixed link}} = \frac{1}{QRR}$$

1. Kinematic pair $Dof = 1$



Helical pair / H-pair / screw pair

$$S_y = f(\theta_y)$$

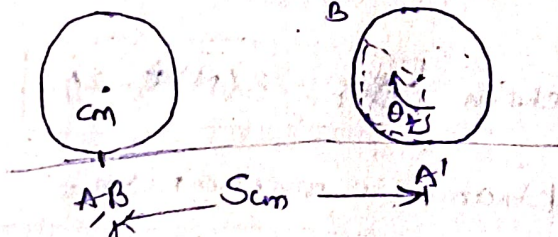
↳ Indep't var

$$\frac{\Delta S_y}{\Delta \theta_y} = \frac{\text{lead}}{2\pi}$$

no. of start
Lead = $n \times \text{pitch}$

$$\boxed{Dof = 1}$$

Higher pair



For pure rolling $A A' = B B'$

$$\boxed{Dof = 1}$$

$$\text{dep } S_{C1C2} = r_1 \theta_1 + r_2 \theta_2 \rightarrow \text{Indep}$$

For Rolling + Slipping

$$\boxed{Dof = 2}$$

$$S_{C1C2} \neq r_1 \theta_1 + r_2 \theta_2 \rightarrow \text{Indep}$$

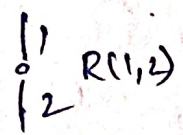
Summary back side start not this start
 No. of links $\rightarrow$ No. of joints $\rightarrow$ No. of higher pairs $\rightarrow$ rolling + slipping

$$\text{DOF} = 3(N-1) - 2j - h$$

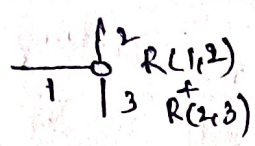
$$\text{DOF} = 3(N-1) - 2j - 2h$$

$\rightarrow$ pure rolling

Binary pair



Ternary pair



If no. of links connected together

$$\text{Joints} = n - 1$$

If In m/c no link fixed $\rightarrow$ chain

$$\text{DOF} = 3(N) - 2j$$

Link fixed $\rightarrow$ kinematic mechanism =

$$\text{DOF} = 3(N-1) - 2j - h$$

DOF < 1 Super structure / redundant structure

DOF = 0 Truss

DOF = 1 Kinematic mechanism

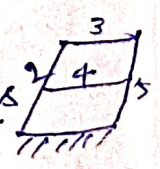
DOF > 1 Unconstrained mechanism

N_r = No. of redundant links
 j_r = No. of redundant joints
 F_r = no. of redundant DOF

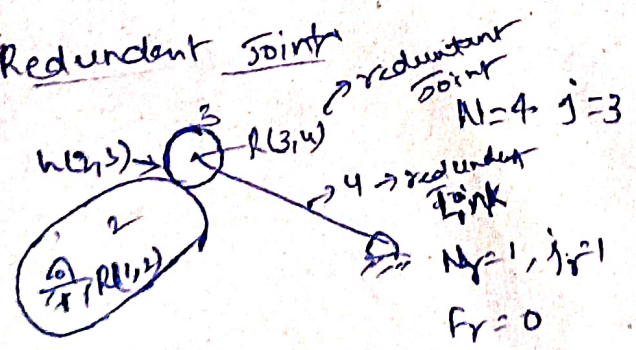
Modified Kutzbach eqn

$$\text{DOF} = 3(N - N_r - 1) - 2(j - j_r) - h - F_r$$

Redundant joints are always 11e to the any of the links



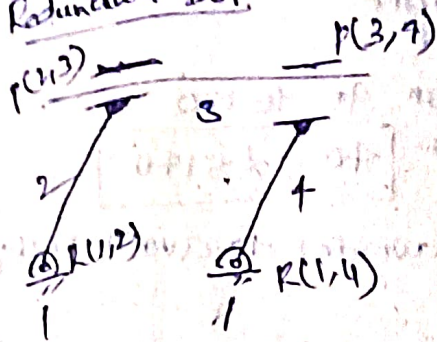
Redundant joints



$$\text{DOF} = 3(N - N_r - 1) - 2(j - j_r) - h - F_r$$

$$= 3(4 - 1 - 1) - 2(3 - 1) - 1 - 0 = 1$$

Redundant DOF



$$N=4, S=4, fr=1$$

$$N_0=0, r=0, h=0$$

$$DOF = 3(4-0-1) - 2(4-0) - 0 - 1 = 9 - 8 - 1 = 0$$

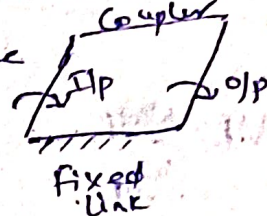
• When two gear are Meshed always Sliding + Rolling contact only

Class-I

Grashof law

$$l_{min} + l_{max} < P + q$$

class-I linkage



Invr-1

Double crank mechanism

Invr-2

Double rocker m/c crank

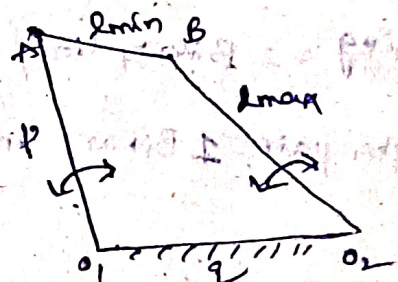
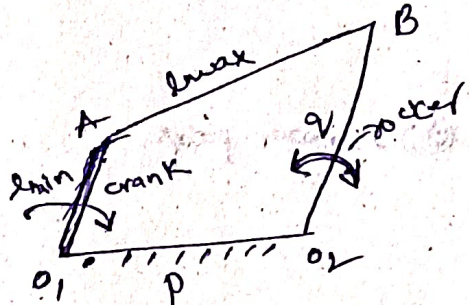
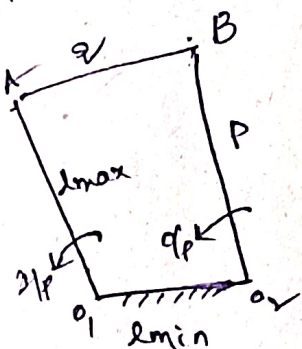
Invr-3

Double rocker m/c

Fixed link = l_{min}

Fixed link: adj to l_{min}

Fixed link = link opp to l_{min}



Locus A = Circle
B = Circle

Locus A = circle
B = Arc of circle

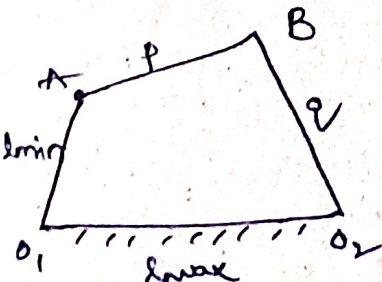
Locus A & B = arc of circle

Class-II

Non-grashof

$$l_{min} + l_{max} > P + q$$

Fixed link = $l_{min} / l_{max} / P / q$

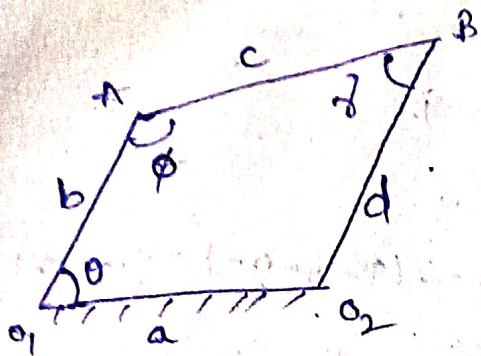


Locus of A & B are arc of circle

Invr: Rocker + Rocker / Triple rocker mechanism

Class-III

$$l_{min} + l_{max} = P + q$$



ϕ = pressure angle

δ = Transmission Angle 40° to 140°

$$\text{Angle } 40^\circ \leq \delta \leq 140^\circ$$

δ = Small values avoided to avoid locking

Mechanical adv $M.A = \frac{\text{load}}{\text{effort}} > 1$

$$M.A = \frac{\sin \delta}{\sin \phi}$$

$$V.R = \frac{\omega_{\text{eff}}}{\omega_{\text{load}}}$$

$$\eta = (M.A) \cdot (V.R) \quad (M.A > 1) \ \& \ (V.R < 1)$$

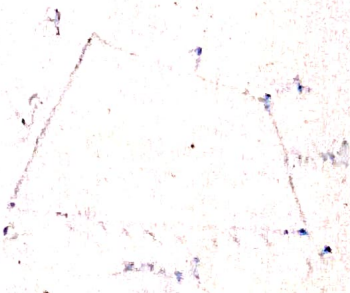
Velocity ratio

M.A

$$\eta = \left(\frac{M.A}{V.R} \right) \quad (V.R > 1)$$

Spring = 2 Binary links

1 higher pair = 1 Binary link & 2 lower pairs



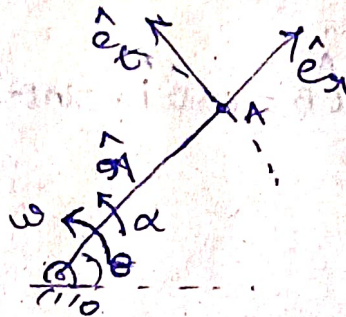
$$L + P = 2n - 2$$

Summary from here, dg 23/10/15

velocity $\vec{V} = \dot{x}\hat{e}_x + y\dot{\theta}\hat{e}_t$

↓
vel of
point along
rod OA

↓
vel of A
w.r.t O
to the rod



$$\vec{a} = \ddot{x}\hat{e}_x + 2\dot{x}\dot{\theta}\hat{e}_t + \dot{x}\ddot{\theta}\hat{e}_t + \dot{x}\dot{\theta}(-\hat{e}_n)$$

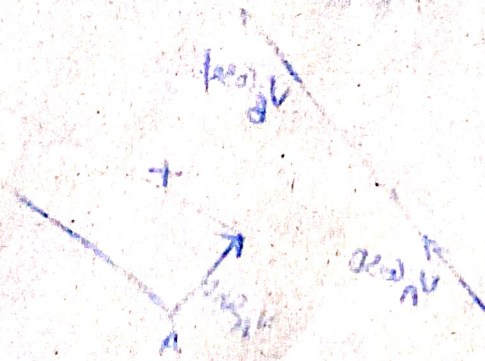
↓
Acc A along
rod

↓
Coriolis
acc $\perp$
to rod

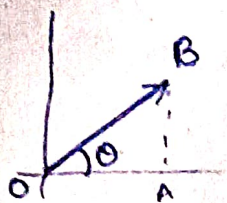
↓
Tang acc
of A w.r.t O
 $\perp$ to rod

↓
Radial acc of A
w.r.t O along the rod
(directed from A towards O)

$$\vec{a}_{\text{Resultant}} = \vec{a}_A + \vec{a}_{\text{Cor}} + \vec{a}_{\text{Tang}} + \vec{a}_{\text{Rad}}$$



Summary

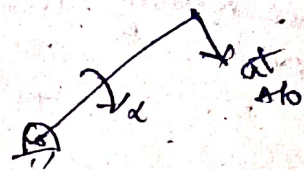
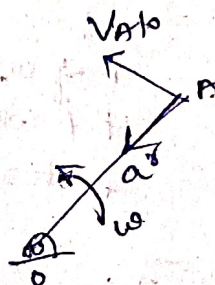
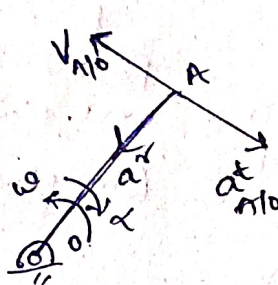
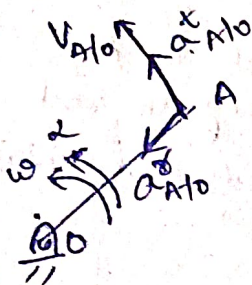


$$\hat{OB} = \cos \theta \hat{i} + \sin \theta \hat{j}$$

Multiplication of velocity

$$\begin{aligned} \hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} &= 1 \\ \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} &= 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} \hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} \\ \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} \end{aligned}} \right\} \text{Dot product}$$

Velocity and acc Analysis of single link



$\omega = \text{const}, \alpha = 0$

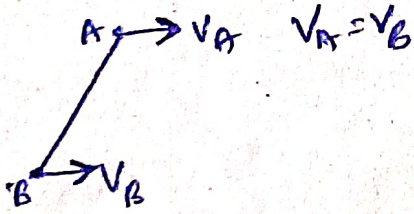
$\omega \neq 0, \alpha \neq 0$

Start from rest / max rotation

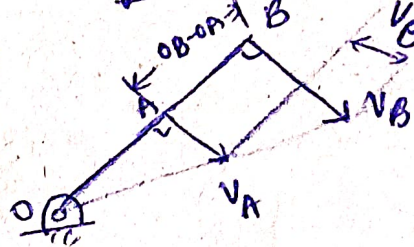
Cylinder disc $\rightarrow$ pure rolling $\rightarrow$ point of contact $\rightarrow$ velocity = 0

Recognition of motion

Translation



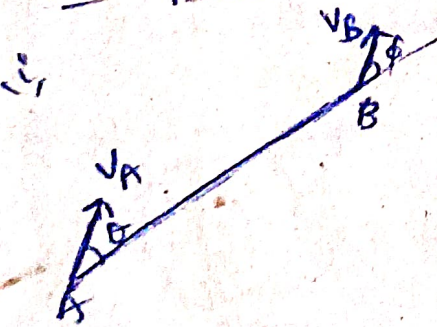
Angular



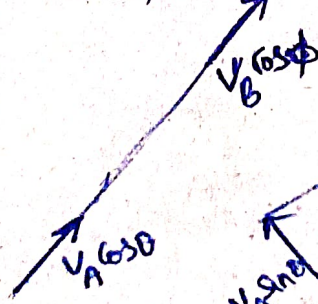
$$\frac{V_B}{OB} = \frac{V_A}{OA} = \frac{V_B - V_A}{OB - OA}$$

$$\frac{V_B}{OB} = \frac{V_A}{OA} = \frac{V_{B/A}}{AB}$$

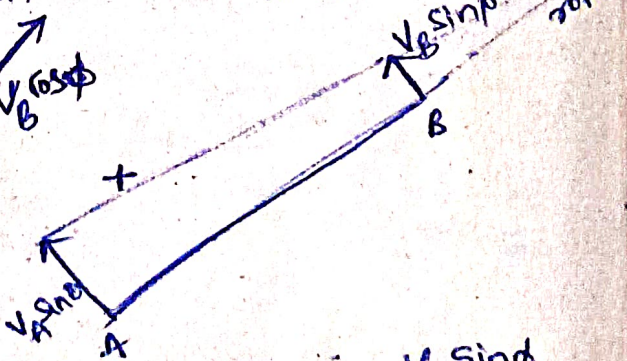
General plane motion



Translation



Rotation

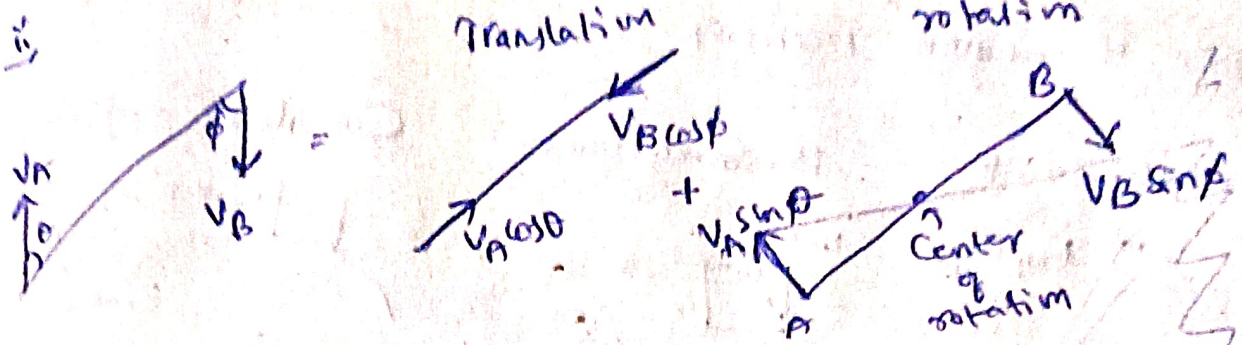


$$V_A \cos \phi = V_B \cos \phi$$

$$\omega_{AB} = \frac{V_A \sin \phi - V_B \sin \phi}{AB}$$

$V_A = V_B$
 $\theta = \phi$ } pure translation

Translation > rotation



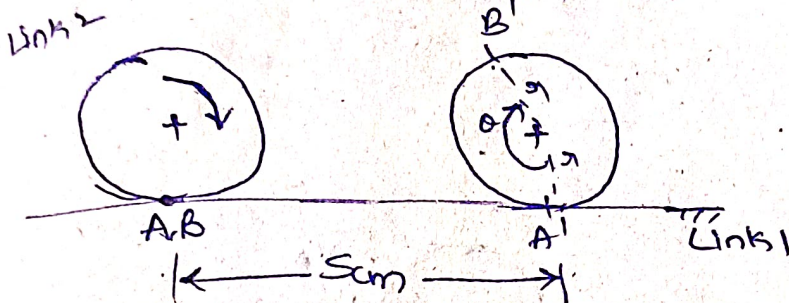
$V_A = V_B$ } pure rotation
 $\theta = \phi$

rotation > translation

$$V_A \cos \theta = V_B \cos \phi$$

$$\omega_{AB} = \frac{V_A \sin \theta - (-V_B \sin \phi)}{AB}$$

Rolling motion

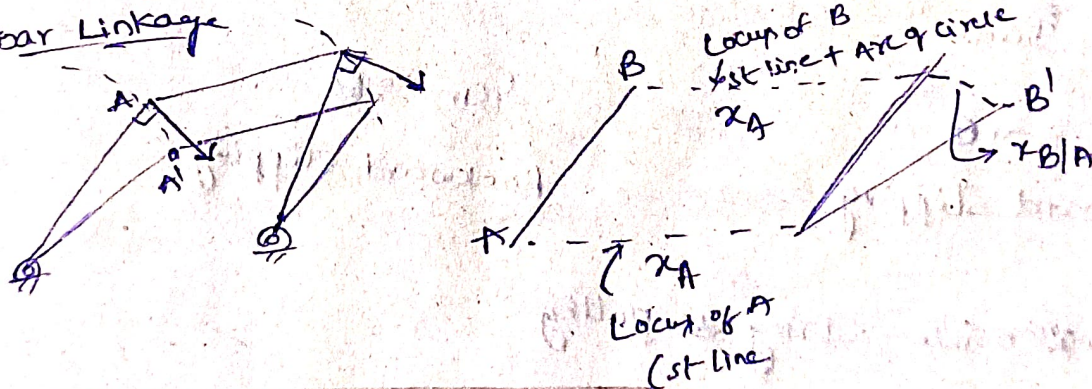


Cond of pure rolling: $\vec{AA'} = \vec{BB'}$
 $S_{cm} = r\theta$

$$V_{cm} = r\omega$$

$$a_{cm} = r\alpha$$

4-bar Linkage



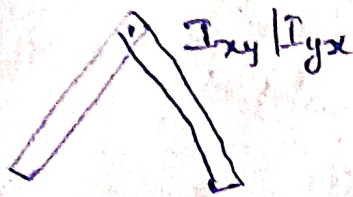
Displacement of B: $x_B = x_A + x_{B/A}$

velocity of B: $\vec{V}_B = \vec{V}_A + \vec{V}_{B/A}$

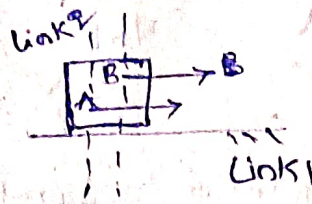
Acceleration of B: $\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$

Some of Common I-centres

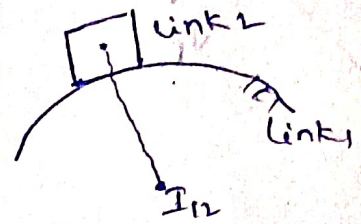
Revolute Joint



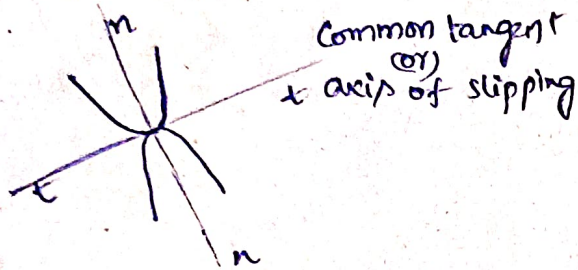
Prismatic Joint



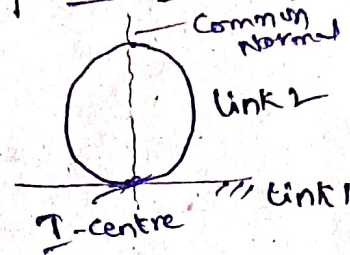
Curvilinear motion



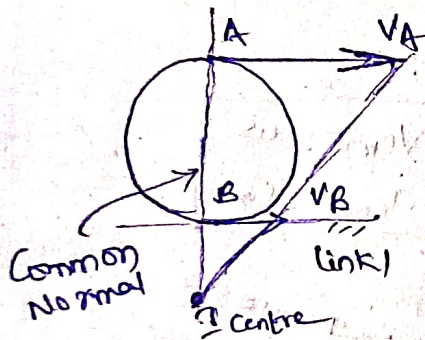
Higher pair



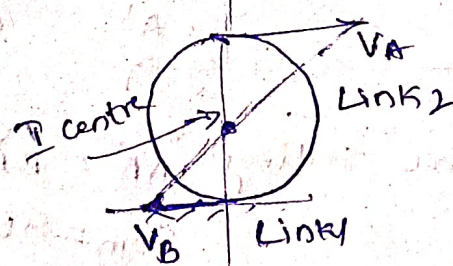
pure rolling



Rolling + Slipping

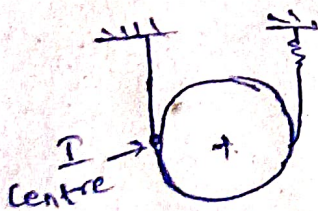


$S_{cm} > \pi \odot$
Forward Slipping



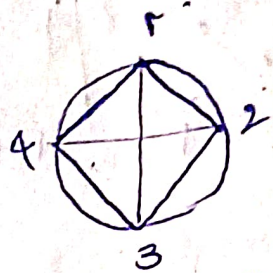
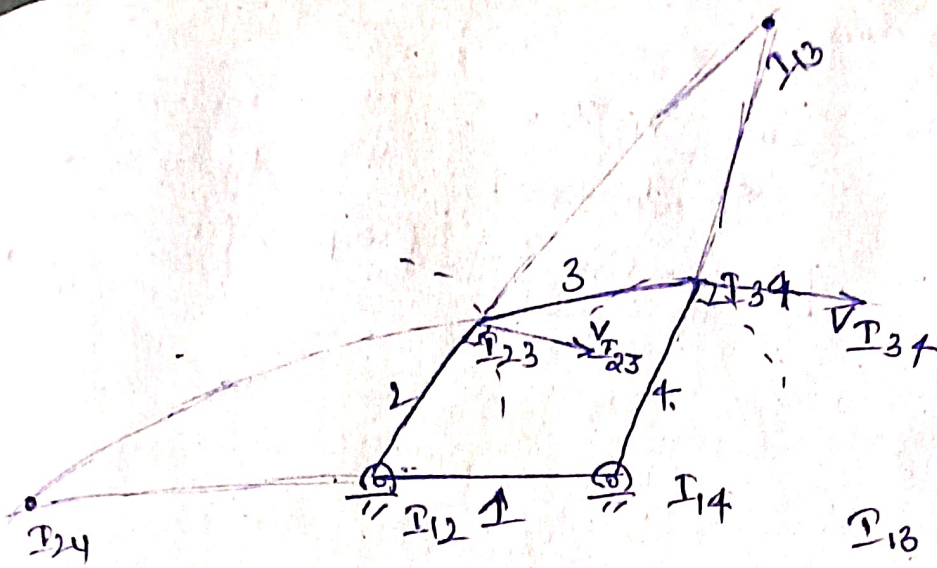
$S_{cm} < \pi \odot$
Backward Slipping

2) Inextensible string & pulley



$$\text{No. of I centres} = n_c = \frac{n(n-1)}{2}$$

$n = \text{No. of links}$



I_{13} Colinear $\begin{cases} I_{12}, I_{23} \\ I_{14}, I_{34} \end{cases}$

I_{24} Colinear $\begin{cases} I_{12}, I_{14} \\ I_{23}, I_{34} \end{cases}$

$$V_{I_{xy}} = I_{1x} I_{xy} \omega_x = I_{1y} I_{xy} \omega_y$$

Absolute
I-centre

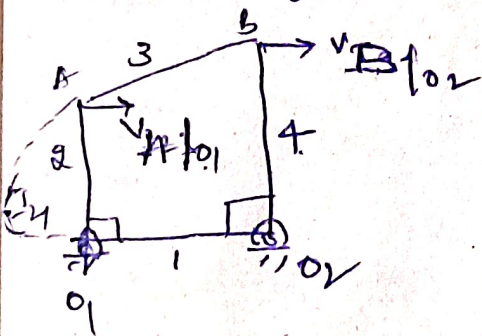
Relative
I-centre

$$V_{I_{23}} = I_{12} I_{23} \omega_2 = I_{13} I_{23} \omega_3$$

$$V_{I_{34}} = I_{13} I_{34} \omega_3 = I_{14} I_{34} \omega_4$$

$$V_{I_{24}} = I_{12} I_{24} \omega_2 = I_{14} I_{24} \omega_4$$

Imp M
Case 1 I/p & o/p are $\perp$ to each other
Link



$$V_{A/O_1} = V_{B/O_2}$$

$$O_1 A \omega_{O_1 A} = O_2 B \omega_{O_2 B}$$

$$l_2 \omega_2 = l_4 \omega_4$$

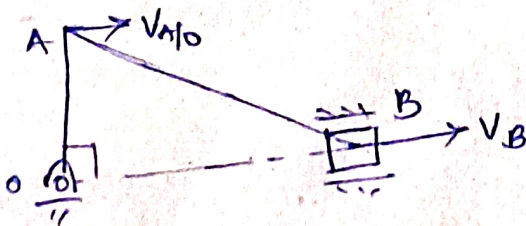
$$\frac{\omega_2}{\omega_4} = \frac{I_{14} I_{24}}{I_{12} I_{24}}$$

Analysis of crank slider me

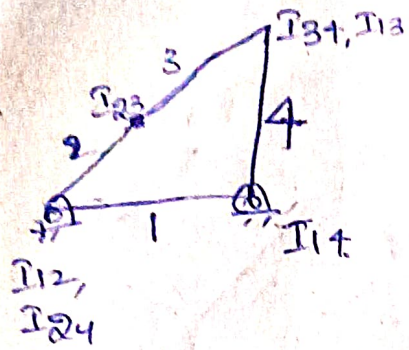
$$V_{A/O} = V_{B/Slider}$$

$$\omega_{AB/O} = 0$$

$$slw = V_{slider}$$



Capee: Toggle position



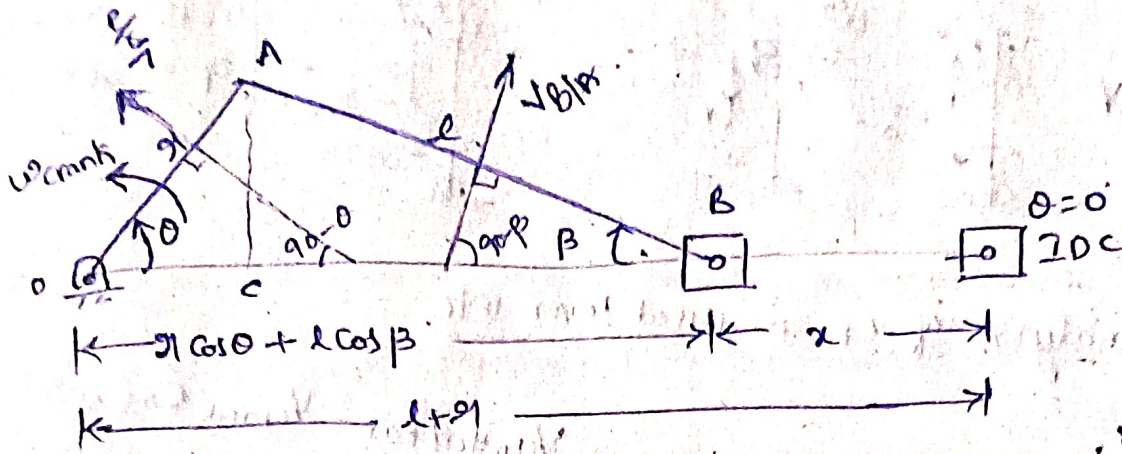
$$\frac{\omega_2}{\omega_3} = \frac{I_{13} I_{23}}{I_{12} I_{23}}$$

$$I_2 \omega_2 = I_3 \omega_3$$

$$\frac{\omega_2}{\omega_4} = \frac{I_{14} I_{24}}{I_{12} I_{24}} = 0$$

$$\boxed{\omega_4 = 0}$$

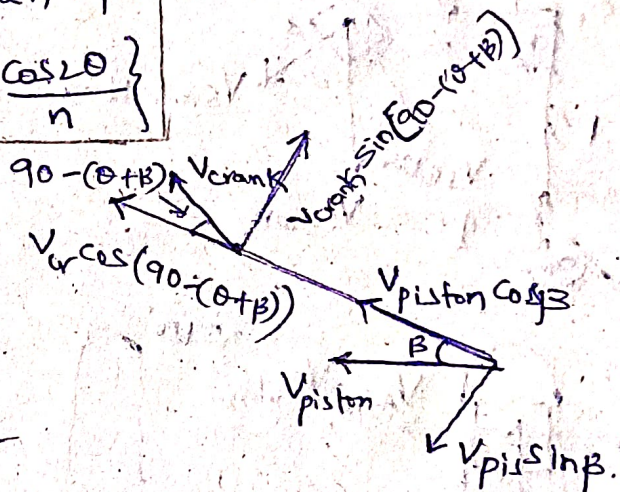
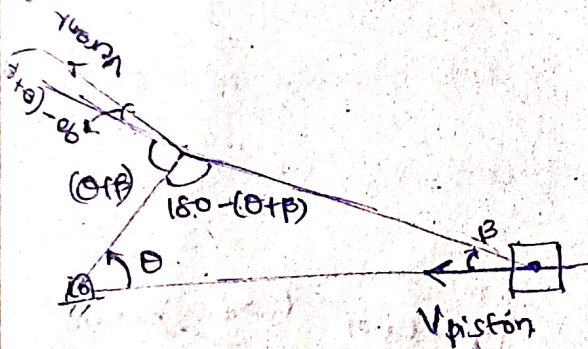
Approximate Analysis of Crank Slider mechanism



$$V_{\text{piston}} = r\omega \left\{ \sin\theta + \frac{\sin 2\theta}{2n} \right\}$$

$$a_{\text{piston}} = r\omega^2 \left\{ \cos\theta + \frac{\cos 2\theta}{n} \right\}$$

$n = \frac{l}{r}$
 $\theta = 0^\circ, V_{\text{piston}} = 0, a_{\text{ps}} = r\omega^2 \left[1 + \frac{1}{n} \right]$



Translation

$$V_{\text{crank}} \cos (90 - (\theta + \beta)) = V_{\text{piston}} \cos \beta \Rightarrow V_{\text{crank}} \sin (\theta + \beta) = V_{\text{piston}} \cos \beta$$

Rotation

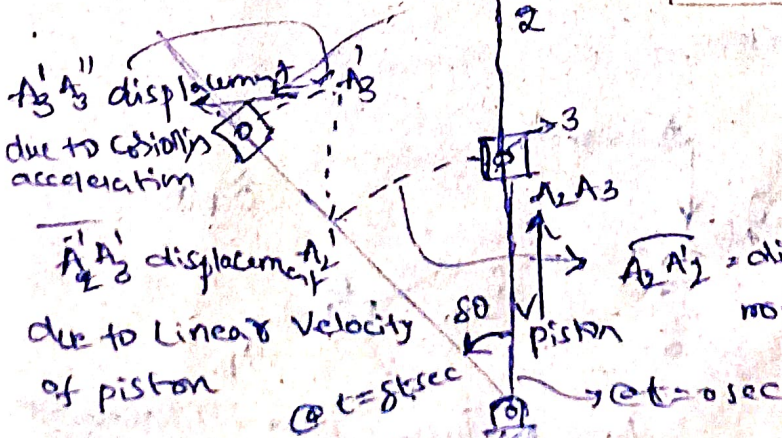
$$\omega = \frac{V_{\text{crank}} \sin (90 - (\theta + \beta)) - (-V_{\text{piston}} \sin \beta)}{AB}$$

Acceleration Analysis

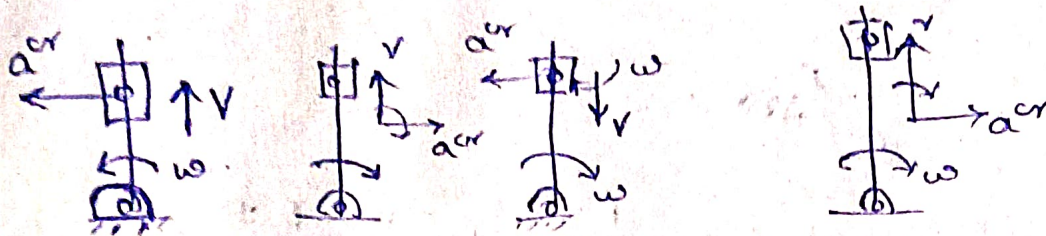
$$a_{\text{cr}} = a_{\text{cr}} + a_{\text{cc}}$$

displacement of piston

$$\widehat{A_2 A_2'} + \widehat{A_2 A_3'} + \widehat{A_3 A_3''}$$



$$\vec{a}_{cr} = 2\dot{\theta}(\hat{e}_\theta) = 2(\omega_{rod} \times V_{slider/rod})$$

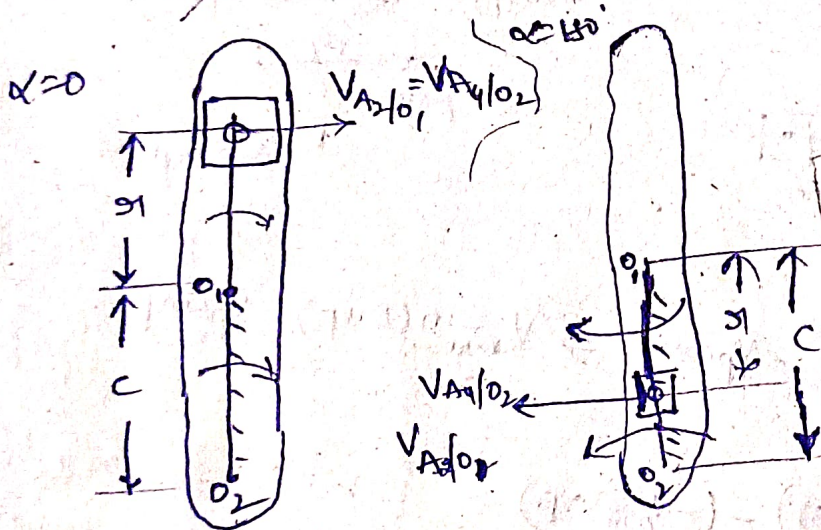
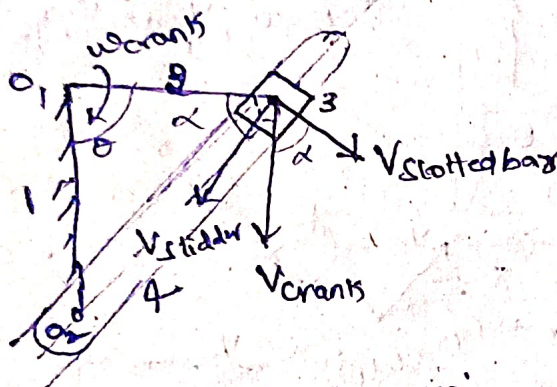


Rotate $V_{slider/rod}$ in the direction of ω_{rod} by 90°

Velocity Analysis of Crank slotted lever m/c

$$V_{slotted\ bar} = V_{crank} \cos \alpha$$

$$V_{slider} = V_{crank} \sin \alpha$$



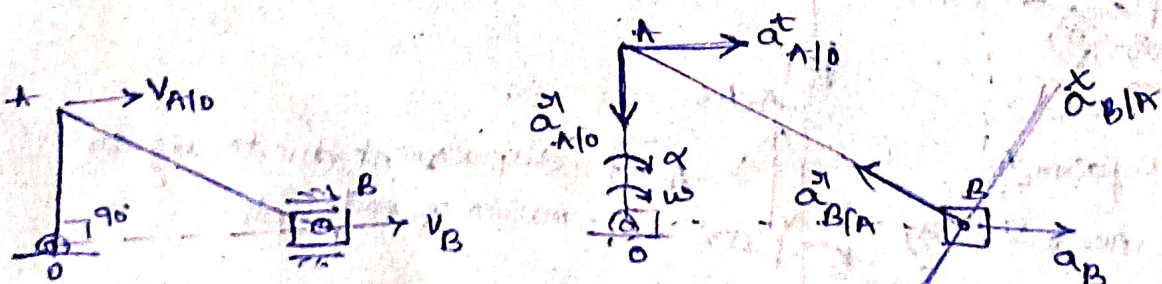
$$\frac{A}{B} = \frac{c+\gamma}{c-\gamma} = \frac{(\omega_{max})_{return}}{(\omega_{max})_{cutting}}$$

$$V_{A2/O1} = V_{A4/O2}$$

$$V_{A1/O1} = V_{A4/O2}$$

$$\omega_{crank} = (c+\gamma)(\omega_{sb})_{max1} \rightarrow \textcircled{A} \quad \omega_{crank} = (c-\gamma)(\omega_{sb})_{max2} \rightarrow \textcircled{B}$$

Acc Analysis of Crank Slider m/c



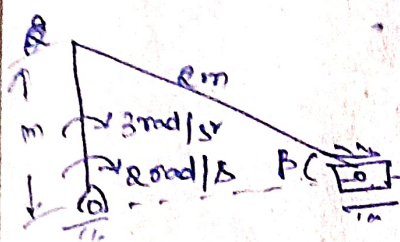
velocity analysis

Acc Analysis

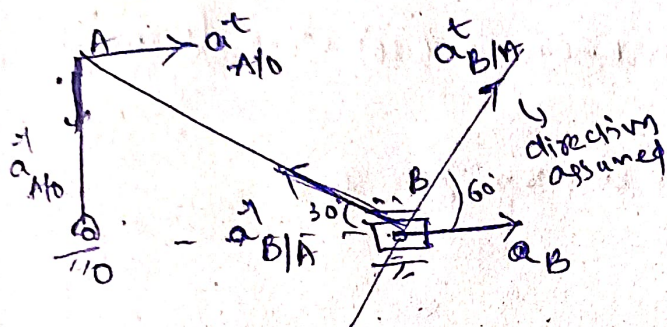
Summary continue after
Vibration chapter

Summary Continuation of summary Velocity and Acc Analysis

acc analysis of Crank slider m/c



acc analysis ↓



$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$$

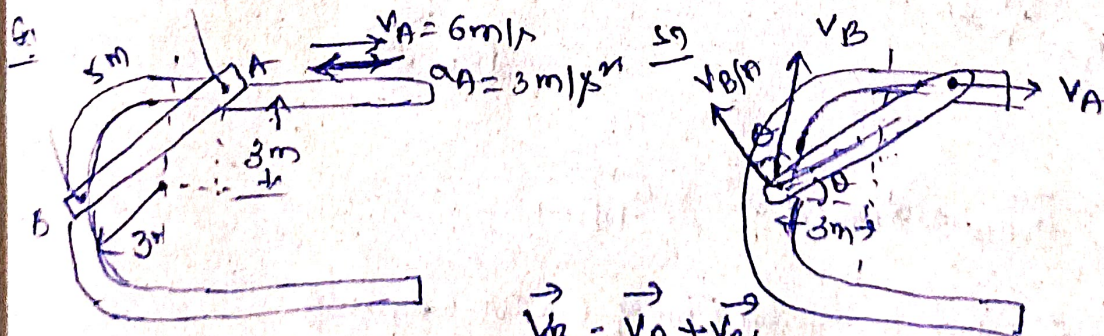
$$\vec{a}_B(\hat{i}) = \vec{a}_{A/O} + \vec{a}_{A/O}^t + \vec{a}_{B/A} + \vec{a}_{B/A}^t$$

$$\vec{a}_B(\hat{i}) = OA\omega^2(-\hat{j}) + OA\alpha\hat{i} + AB\omega_{AB}^2(-i\cos 30 + j\sin 30) + AB\alpha_{AB}(i\cos 60 + j\sin 60)$$

$$\vec{a}_B(\hat{i}) = 1 \times 2^2(-\hat{j}) + 1 \times 3\hat{i} + 2 \times 0^2(-i\cos 30 + j\sin 30) + 2 \times \alpha_{AB}(i\cos 60 + j\sin 60)$$

$$\text{Coeff } \hat{j} = 0 = -4 + 2\alpha_{AB}\sin 60 \Rightarrow \alpha_{AB} = \frac{2}{\sin 60} \text{ rad/s}^2$$

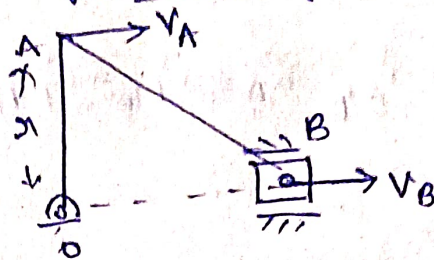
$$\text{Coeff } \hat{i} = a_B = 3 + 2\alpha_{AB}\cos 60 = 3 + 2 \times \frac{4}{\sqrt{3}} \cos 60 = 4.2 \text{ m/s}^2$$



$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$$

$$\vec{v}_B(\hat{j}) = 6(\hat{i}) + AB\omega_{AB}(-i\sin \theta + j\cos \theta)$$

velocity analysis



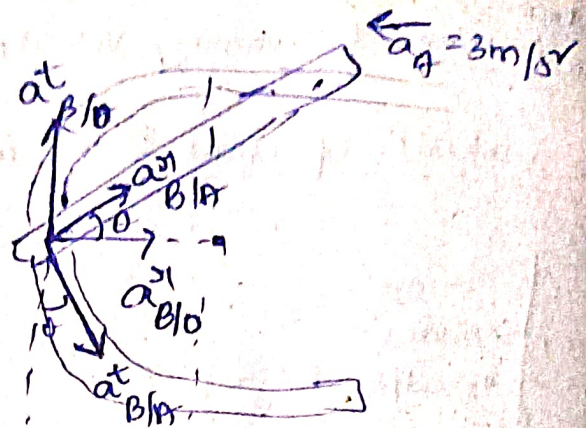
$$v_A = v_B$$

$$\omega \times 1 = v_B \Rightarrow v_B = 1 \times 2 = 2 \text{ m/s}$$

Acc Analysis

$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$$

$$\vec{a}_{B/O}^n + \vec{a}_{B/O}^t = \vec{a}_A + \vec{a}_{B/A}^n + \vec{a}_{B/A}^t$$



$$\frac{(V_B)^n}{OB} \hat{i} + OB \alpha_{OB} = 3(-\hat{i}) + AB \omega_{AB}^n (\cos 36.86 + j \sin 36.86) + AB \alpha_{AB} (i \sin 36.86 - j \cos 36.86)$$

$$\frac{8}{3} \hat{i} + 3 \alpha_{OB} = -3 \hat{i} + 5 \times 2 (\cos 36.86 + j \sin 36.86) + 5 \alpha_{AB} (i \sin 36.86 - j \cos 36.86)$$

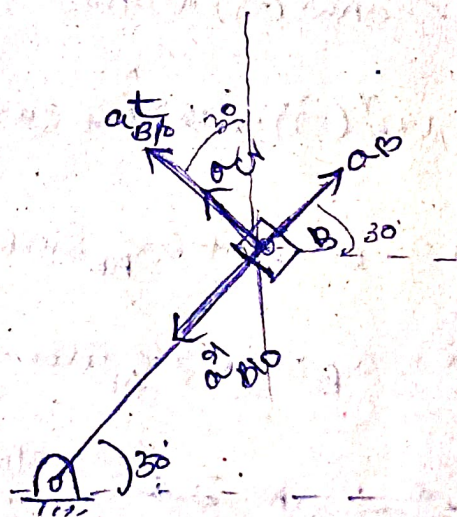
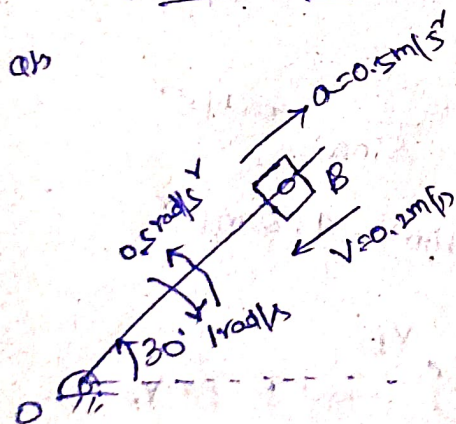
$$i \text{ coeff } \frac{8}{3} = -3 + 5 \times 2 \cos 36.86 + 5 \alpha_{AB} \sin 36.86$$

$$\alpha_{AB} = 2.78 \text{ rad/s}$$

$$j \text{ coeff } 3 \alpha_{OB} = 2 \sin 36.86 - 5 \alpha_{AB} \cos 36.86 \quad \alpha_{OB} = 0.28 \frac{\text{rad}}{\text{s}}$$

$$\text{Acc at Point B} = \sqrt{(a_{B/O}^n)^2 + (a_{B/O}^t)^2} = \sqrt{\left(\frac{8}{3}\right)^2 + (3 \times 0.28)^2} = 2.4 \text{ m/s}^2$$

Ex: Grills acc prob



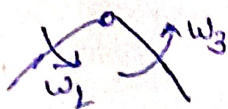
$$\vec{a}_{\text{result}} = \vec{a}_{B/O}^n + \vec{a}_{B/O}^t + \vec{a}_B^c + \vec{a}_{B/\text{rod}}$$

$$= OB \omega_{OB}^n (-i \cos 30 - j \sin 30) + OB \alpha_{OB} (-i \sin 30 + j \cos 30) + 2v_{\text{slide}} \omega_{\text{rod}} (-i \sin 30 + j \cos 30) + a_B (i \cos 30 + j \sin 30)$$

$$\vec{a}_{\text{res}} = -0.213 \hat{i} + 0.565 \hat{j} \Rightarrow a_{\text{res}} = \sqrt{0.213^2 + 0.565^2} = 0.61 \text{ m/s}^2$$

$$\text{dire w.r.t x axis } \theta = \tan^{-1} \left(\frac{j \text{ coeff}}{i \text{ coeff}} \right) = \tan^{-1} \left(\frac{0.565}{-0.213} \right) = -69.4^\circ = 180 - 69.4 = 110.6^\circ$$

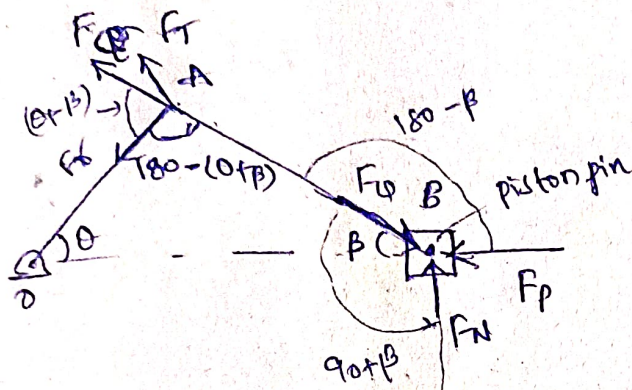
Slipping velocity



$$V_{\text{slipping}} = r_{\text{pin}} (\omega_2 - \omega_3)$$

$$V_{\text{slub}} = r_{\text{pin}} [\omega_2 - (-\omega_3)]$$

Dynamic Analysis



$$\frac{F_Q}{\sin 90} = \frac{F_P}{\sin(90+\beta)} = \frac{F_N}{\sin(180-\beta)}$$

$$\Rightarrow F_Q = \frac{F_P}{\cos \beta}$$

$$F_T = F_Q \sin(\theta + \beta)$$

$$F_R = F_Q \cos(\theta + \beta)$$

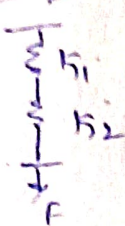
$$\text{Torque } T = F_T \times r$$

Vibration short notes (not full)

axial stiffness = $\frac{AE}{L}$ axial rigidity

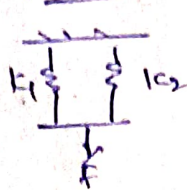
Torsional stiffness = $\frac{GJ}{L}$ Torsional rigidity

Series



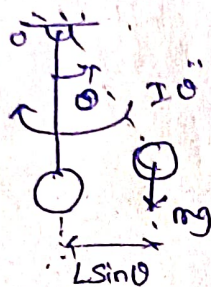
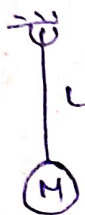
$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

Parallel



$$k_{eq} = k_1 + k_2$$

Simple pendulum



$$\sum M = 0$$

$$I\ddot{\theta} + MgL \sin \theta = 0$$

$$M\ddot{\theta} + MgL \theta = 0$$

$$\ddot{\theta} + \frac{g}{L} \theta = 0$$

$$\omega_n = \sqrt{\frac{g}{L}}$$

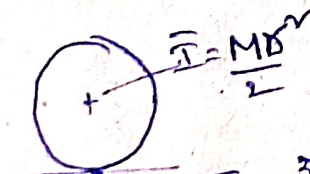
Time period = $t_n = \frac{2\pi}{\omega_n}$ (sec)

$\omega_n = \sqrt{\frac{k}{M}}$ stiffness $\frac{\text{rad}}{\text{r}}$

$\downarrow$ Natural freq

frequency $f = \frac{1}{t_n}$ Hz

Gynderal disc



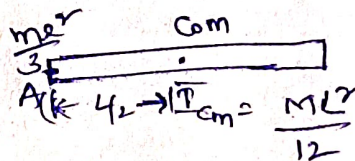
$$I_{poc} = \frac{3}{2} M r^2$$

$$I_{poc} = I_{cm} + m d^2$$

$$= \frac{M r^2}{2} + M r^2$$

$$= \frac{3}{2} M r^2$$

Slender rod

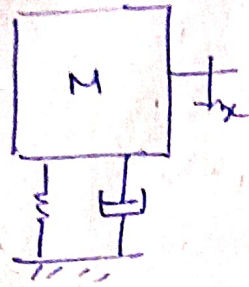


$$I_A = I + m d^2$$

$$= \frac{M L^2}{12} + M \left(\frac{L}{2}\right)^2$$

$$I_A = \frac{M L^2}{3}$$

Free damped vib



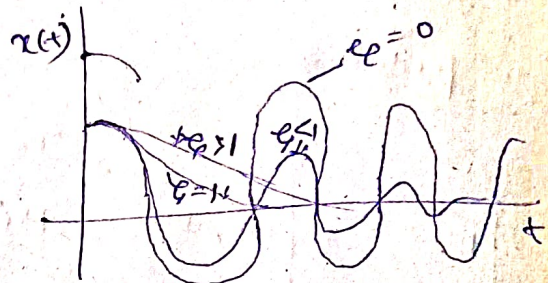
$$M\ddot{x} + c\dot{x} + kx = 0$$

$\zeta < 1$ underdamped

$\zeta = 0$ undamped

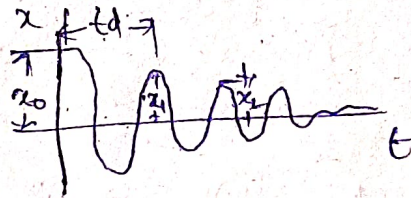
$\zeta = 1$ critical damped

$\zeta > 1$ overdamped



Logarithmic decrement

$$\frac{1}{n} \ln \frac{x_0}{x_n} = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$$



Damping ratio
(or)
damping factor

$$\zeta = \frac{c}{c_c} = \frac{c}{2\sqrt{mk}}$$

critical
damping coeff

Motion

Eq of motion

$$M\ddot{x} + c\dot{x} + kx = 0$$

Natural freq

$$\omega_n = \sqrt{\frac{k}{m}}$$

Critical damping

$$c_c = 2\sqrt{mk}$$

damping ratio

$$\zeta = \frac{c}{2\sqrt{mk}}$$

Rotation

$$I_{eq}\ddot{\theta} + c_{eq}\dot{\theta} + k_{eq}\theta = 0$$

$$\omega_n = \frac{k_{eq}}{I_{eq}}$$

$$c_c = 2\sqrt{I_{eq}k_{eq}}$$

$$\zeta = \frac{c}{2\sqrt{I_{eq}k_{eq}}}$$

Parameter	Motion			
	Translation		Rotation	
	Force	Energy	Couple	Energy
Inertia	$M\ddot{x}$	$\frac{1}{2}m\dot{x}^2$	$I_{eq}\ddot{\theta}$	$\frac{1}{2}I_{eq}\dot{\theta}^2$
Restoration	kx	$\frac{1}{2}kx^2$	$k_{eq}\theta$	$\frac{1}{2}k_{eq}\theta^2$
Damping	$c\dot{x} = F_D$	$\int F_D dx = \int c\dot{x} dx$	$c_{eq}\dot{\theta}$	$\int c_{eq}\dot{\theta} d\theta$

Damped freq $\omega_d = \omega_n \sqrt{1 - \zeta^2}$

$$\frac{I_{eq}}{\text{point}} \ddot{\theta} + c \left(\frac{\pm \text{er dist from ref point}}{\text{point}} \right) \dot{\theta} + k \left(\frac{\pm \text{er dist from ref point}}{\text{point}} \right) \theta \pm mg \left(\frac{\pm \text{er dist from ref point}}{\text{point}} \right) \theta = 0$$

* V.B.P *

Forced vibration

$$(DMF) = \frac{\text{Amplitude of Forced vib}}{\text{Static deflection}}$$

Dynamic Magnification
factor

$$DMF = \frac{A}{\delta_{static}}$$

$$A = \frac{F}{\sqrt{(k - M\omega^2)^2 + (c\omega)^2}}$$

$$DMF = \frac{x_{max}}{x_{static}} = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}$$

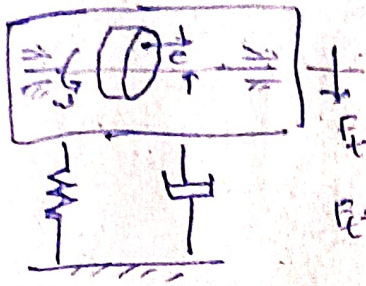
$$\delta_{static} = \frac{F}{k}$$

Force transmitted to Foundation

$$\epsilon = \frac{F_T}{F_A} = \frac{k^2 + (c\omega)^2}{(k - M\omega^2)^2 + (c\omega)^2}$$

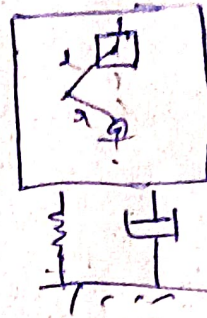
Transmissibility ratio

Excitation due to unbalanced rotating mass



$$F_x = me\omega^2 \sin \theta$$

$$F_x = me\omega^2 \sin(\omega t)$$

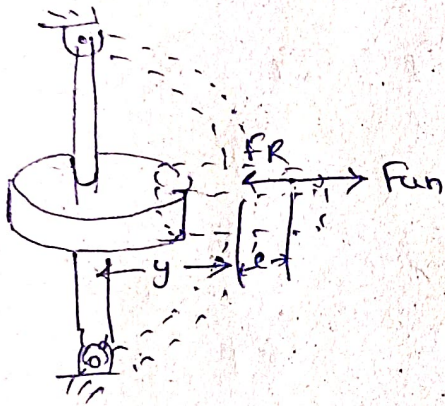


$$F(t) = -m\pi\omega^2 \cos \theta$$

Amplitude $x = \frac{me\omega^2}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$

$$A = \frac{m\pi\omega^2}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

Whirling of shaft



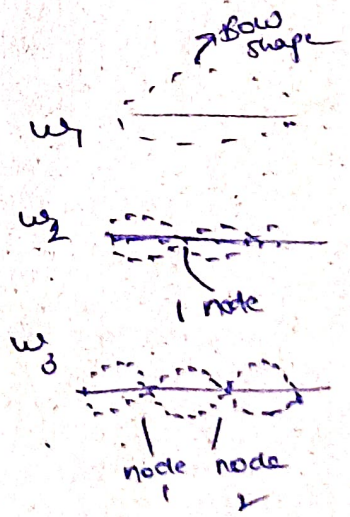
$$y = \frac{e\omega^2}{(1 - r^2)}$$

$$r = \frac{\omega}{\omega_n}$$

Mode shapes

$$\omega = \omega_1, 2\omega_1, 3\omega_1, \dots, n\omega_1$$

$$\omega = \omega_1, \omega_2, \omega_3, \dots$$



Gears & Gear trains

Tom
3 Faculty

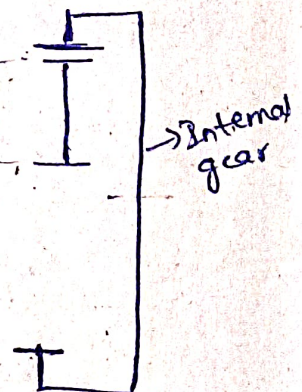
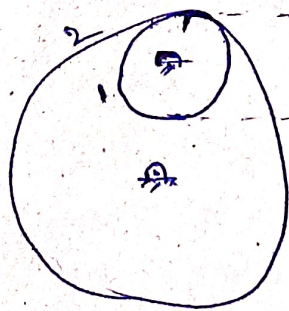
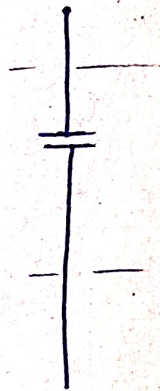
1. Velocity ratio = $\frac{d}{d}$

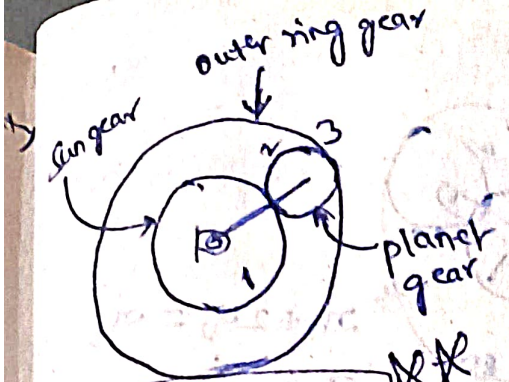
3. $m = \frac{d}{T}$ (module), diametric pitch = $\frac{T}{d}$, circular pitch (p_c) = $\frac{\pi D}{T}$

10) $\text{Power} = \text{Torque} \times \omega$
 $\text{Torque} = F_T \times \frac{d}{2}$ (Tangential force)
 $\frac{F_R}{F_T} = \tan \phi$ (Radial force)
 ϕ = pressure gear angle

11. $\frac{N_{\text{driven}}}{N_{\text{driver}}} = \frac{\sum (T_{\text{driver}})}{\sum (T_{\text{driven}})}$

$\frac{N_2}{N_1} = \frac{t_1}{t_2} = \frac{\text{Torque}_1}{\text{Torque}_2}$





$$N_{arm} = y$$

$$N_1 = y + x$$

$$N_2 = y - x \left[\frac{T_1}{T_2} \right]$$

$$N_3 = y - x \left[\frac{T_1 \times T_2}{T_2 \times T_3} \right] = 0$$

$$y - x \left[\frac{1}{2} \right] = 0$$

$$2y - x = 0$$

$$\frac{d_1}{2} + 2 \frac{d_2}{2} = \frac{d_3}{2}$$

$$\frac{m T_1}{2} + m T_2 = \frac{m T_3}{2}$$

$$m \left\{ \frac{T_1}{2} + T_2 \right\} = m T_3$$

$$m = \frac{d}{T}$$

$$2y - x = 0$$

$$y + x = 100$$

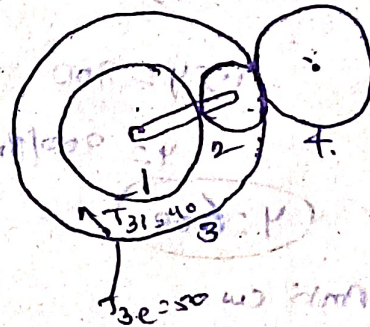
$$3) N_{arm} = y$$

$$N_1 = y + x$$

$$N_2 = y - x \left[\frac{T_1}{T_2} \right]$$

$$N_3 = y - x \left[\frac{T_1}{T_2} \times \frac{T_2}{T_3} \right]$$

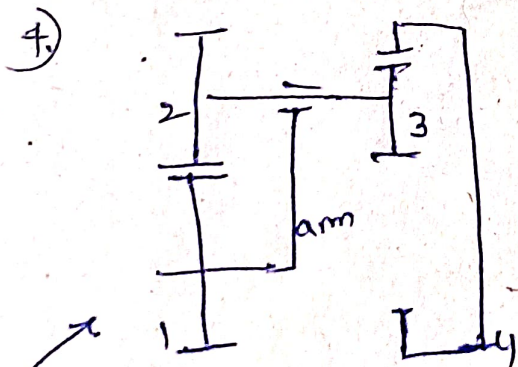
$$N_3 = y + x \left[\frac{T_1}{T_2} \times \frac{T_2}{T_3} \times \frac{T_3}{T_4} \right]$$



driver
driven

$$\frac{T_1}{d_1} + \frac{T_2}{d_2} + \frac{T_3}{d_3} = \frac{T_4}{d_4}$$

driver
driven



$$N_{arm} = y$$

$$N_1 = y + x$$

$$N_2/N_3 = y - x \left[\frac{T_1}{T_2} \right] \quad (N_2 = N_3)$$

$$N_2/N_3$$

$$N_4 = y - x \left[\frac{T_1}{T_2} \times \frac{T_3}{T_4} \right]$$

see carefully

$\omega = -ve$
 $ccw = +ve$

PHY practice by me

21. Normal

$$N_{arm} = y = 9$$

$$N_{gear} = y + x = 0 \quad x = -y$$

$$N_2 = y - x \left(\frac{T_1}{T_2} \right)$$

$$N_3 = y - x \left(\frac{T_1}{T_2} \times \frac{T_2}{T_3} \right) = 900$$

$$= y - (-y) \left[\frac{40}{20} \times \frac{20}{80} \right] = 900$$

$$y + y \left(\frac{1}{2} \right) = 900$$

$$1.5y = 900$$

$$y = 900 / 1.5$$

$$y = 600$$

20) Anti cw

$$\frac{N_2}{N_1} = \frac{T_2}{T_1} \times \frac{T_4}{T_5} \times \frac{T_5}{T_6}$$

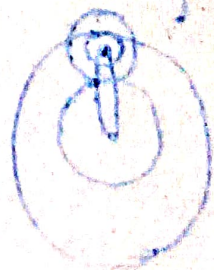
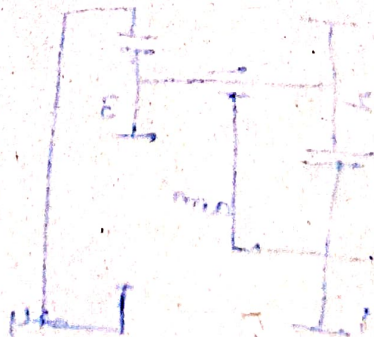
$$\frac{18T}{44T} \times \frac{15T}{33T} \times \frac{33T}{36T}$$

$$\frac{N_2}{N_1} = \frac{D_2}{D_1}$$

$$\frac{1800}{N_1} = \frac{150}{250}$$

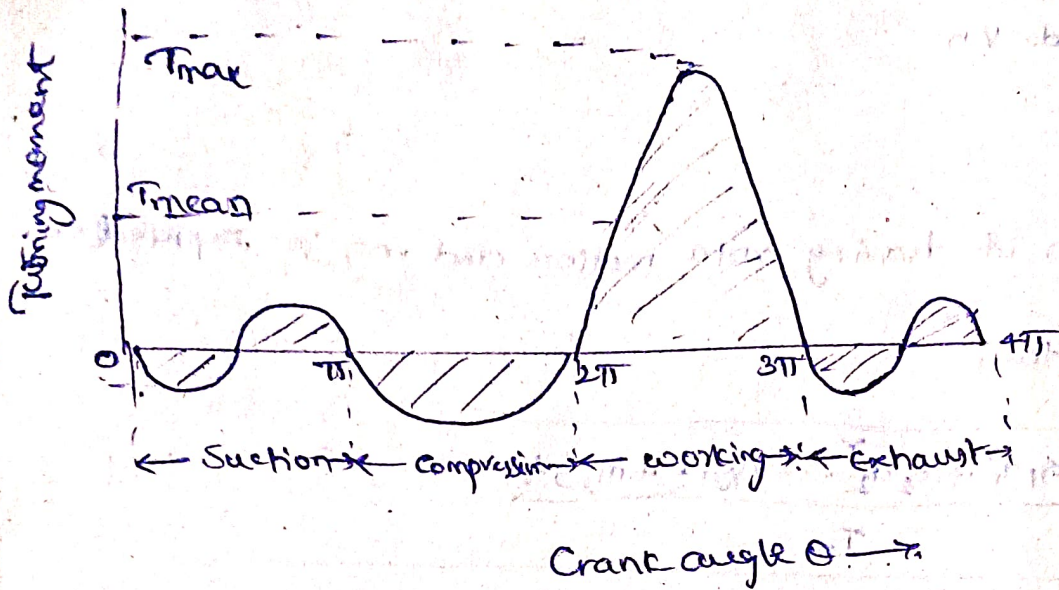
$$\Rightarrow N_1 = \frac{2500}{1.5}$$

$$N_1 = \frac{164}{x + y} = 164$$



Fly wheel

Single cylinder 4-stroke I.C Engine



Cycle time = $2\pi \rightarrow 2$ stroke

Cycle time = $4\pi \rightarrow 4$ stroke

Max Energy stored in flywheel $(\Delta KE)_{max} = I \omega_{avg}^2 \cdot C_s$ → Coeff of fluctuation in speed

Coeff of fluctuation of energy $C_E = \frac{\text{max fluctuation in Energy (or) Energy stored in flywheel}}{\text{Energy Supplied}}$ $C_E = \frac{(\Delta KE)_{max}}{E_{supply}}$

Energy supplied $E_{supply} = T_{mean} \times \text{Cycle time}$
 $2\pi \text{ (or) } 4\pi$

Stress in flywheel $\sigma = \rho v^2$

$$v = \frac{\pi D N}{60}$$

↓
2 stroke engine

↓
4 stroke engine

Trusses
m223-3

- > Truss with redundant member
- < Collapsible truss

Self locking mechanism

$Q_{int} = W \sin(\theta + \phi_s) \quad \phi_s > 0$

$Q_{ext} = W \sin(\theta - \phi_s) \quad \phi_s < 0$



$a = v \frac{dv}{ds} = \frac{v^2 \sin^2 \theta}{2g}$

$\rho_{avg} = \frac{u^2 \sin^2 \theta}{g}$

$\rho = \frac{u^2 \sin^2 \theta}{g}$

$e = \frac{|v_2 - v_1|}{|u_2 - u_1|}$

Plastic impact $e = 0$
 Elastic impact $e = 1$

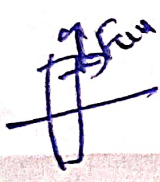
$S = \frac{1}{n} \ln \frac{x_0}{x_1} = \frac{2\pi \eta}{\sqrt{1 - \eta^2}}$

$\rho_{avg} = \frac{A}{k_{static}^2} \frac{1}{\sqrt{(k - m\omega^2)^2 + (2\eta\omega)^2}}$

$A = \frac{F}{\sqrt{(k - m\omega^2)^2 + (2\eta\omega)^2}}$

$e = \frac{x}{y} = \frac{F_T}{F} = \sqrt{\frac{k^2 - (\omega)^2}{(k - m\omega^2)^2 + (\omega)^2}} = \sqrt{\frac{1 + (2\eta\omega)^2}{(1 - \gamma^2)^2 + (2\eta\omega)^2}}$

$\eta = 1 - e$



$F_{ext} = m\omega^2 y \sin \omega t$

$y = \frac{e\gamma^2}{1 - \gamma^2} \approx \frac{\omega}{\omega_n}$



$F_{pis} = m\omega^2 y \cos \omega t$